

Partial Tax Recovery on Returns (α -Reclaim under Seller and Buyer Remittance)

Draft notes
Companion to `welfare_with_returns`

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1 Setup

The companion note studies the extreme in which the remitter recovers none of the unit tax τ from the government when a unit is returned. Here the remitter recovers a fraction $\alpha \in [0, 1)$ of τ on each returned unit, whether the seller or the buyer remits. Full reclaim is the boundary $\alpha = 1$; zero reclaim is $\alpha = 0$. Private refunds between buyer and seller are unchanged from the companion note. Under seller remittance the seller refunds the tax-inclusive checkout price p . Under buyer remittance the seller refunds only the pre-tax list price $p - \tau$, and the buyer reclaims $\alpha\tau$ from the treasury on a return.

Demand $D(p)$, return rate $r(\cdot)$, handling h , and market supply $S(\mu)$ are as in the companion note. Equilibrium clears at $D(p) = S(\mu)$, with μ the remitter- and α -dependent effective margin derived below. Write $Q \equiv D(p)$.

2 Effective margins

Under seller remittance the seller remits τ on every gross sale and recovers $\alpha\tau$ from the government on each return. The private refund remains p , so the return rate is still $r(p)$. Expected net receipt per gross unit is therefore

$$\mu^s(p, \tau; \alpha) = p - \tau - r(p)(p + h - \alpha\tau). \quad (1)$$

The term $-\alpha\tau$ inside the return cost is the treasury reclaim. At $\alpha = 0$ this is the companion-note margin $p - \tau - r(p)(p + h)$.

Under buyer remittance the seller never remits τ . The buyer remits on purchase and recovers $\alpha\tau$ on a return, while the seller refunds only the pre-tax price $p - \tau$. The buyer's total recovery on a return is therefore

$$(p - \tau) + \alpha\tau = p - (1 - \alpha)\tau,$$

so the return rate is evaluated at $p - (1 - \alpha)\tau$. The seller's effective margin is

$$\mu^b(p, \tau; \alpha) = (p - \tau) - r(p - (1 - \alpha)\tau)(p - \tau + h). \quad (2)$$

At $\alpha = 0$ the argument of r collapses to $p - \tau$ and (2) recovers the companion note. Clearing conditions are $D(p) = S(\mu^s(p, \tau; \alpha))$ and $D(p) = S(\mu^b(p, \tau; \alpha))$.

3 Tax neutrality

At a common checkout price p and tax τ , the two regimes define the same equilibrium if and only if $\mu^s = \mu^b$. Canceling $p - \tau$ gives

$$r(p)(p + h - \alpha\tau) = r(p - (1 - \alpha)\tau)(p - \tau + h). \quad (3)$$

Result 1 (Partial recovery and tax neutrality). *Seller and buyer remittance are equivalent if and only if (3) holds. When $\alpha = 1$, both sides equal $r(p)(p - \tau + h)$, so neutrality holds for every r . When $\alpha \in [0, 1)$, neutrality fails for generic r , as in the zero-reclaim companion note.*

Full reclaim undoes the remittance wedge on returns. Any incomplete reclaim leaves a wedge between the seller's return cost and the buyer's keep-or-return price, so which party remits continues to matter.

4 Pass-through

Production-cost and handling pass-through ρ_c and ρ_h are as in the companion note and are unchanged by reclaim. What changes is the tax wedge. Failure of neutrality requires separate checkout prices p^s and p^b . With $S_\tau = 0$,

$$\rho_\tau^s = \frac{-S'(\mu) \mu_\tau^s}{S'(\mu) \mu_p^s - D'(p^s)}, \quad \rho_\tau^b = \frac{-S'(\mu) \mu_\tau^b}{S'(\mu) \mu_p^b - D'(p^b)}. \quad (4)$$

Differentiating (1)–(2) at fixed checkout price, and writing $z \equiv p - (1 - \alpha)\tau$,

$$\mu_p^s = 1 - r(p) - r'(p)(p + h - \alpha\tau), \quad \mu_\tau^s = -(1 - \alpha r(p)), \quad (5)$$

$$\mu_p^b = 1 - r(z) - r'(z)(p - \tau + h), \quad \mu_\tau^b = -(1 - r(z)) + r'(z)(1 - \alpha)(p - \tau + h). \quad (6)$$

Seller remittance lowers μ by the unclaimed expected tax $1 - \alpha r(p)$. Buyer remittance lowers the keep-or-return price by only $1 - \alpha$, so the return-rate channel in μ_τ^b is scaled by that factor.

At $\tau = 0$, both list-price slopes collapse to the companion object $\mu_p \equiv 1 - r(p) - r'(p)(p + h)$, and

$$\mu_\tau^s = -(1 - \alpha r(p)), \quad \mu_\tau^b = -\mu_p - \alpha r'(p)(p + h). \quad (7)$$

Comparing with ρ_c then yields

$$\rho_\tau^s = (1 - \alpha r(p)) \rho_c, \quad \rho_\tau^b = (\mu_p + \alpha r'(p)(p + h)) \rho_c. \quad (8)$$

Result 2 (Tax pass-through under partial reclaim). *At $\tau = 0$, tax pass-through is (8). The formulas nest the companion note at $\alpha = 0$, where $\rho_\tau^s = \rho_c$ and $\rho_\tau^b = \mu_p \rho_c$. They coincide at $(1 - r(p)) \rho_c$ when $\alpha = 1$, so remittance ceases to matter for pass-through exactly when neutrality holds. For $\alpha \in [0, 1)$, $\text{sgn}(\rho_\tau^b) = -\text{sgn}(\rho_\tau^s)$ if and only if $\mu_p + \alpha r'(p)(p + h) < 0$ and $\rho_c \neq 0$.*

How α moves the two tax rates follows from the resource wedge each regime leaves on the supply side. Under seller remittance the firm remits τ on every gross sale and recovers only $\alpha\tau$ on returns, so the expected net tax cost per unit is $(1 - \alpha r)\tau$. A larger reclaim rate shrinks that cost toward the keep-rate burden $(1 - r)\tau$, and therefore scales ρ_τ^s from ρ_c toward $(1 - r)\rho_c$. Cost and handling pass-through are untouched because they never enter the reclaim term.

Under buyer remittance the firm never remits τ , but reclaim changes the buyer's keep-or-return price $z = p - (1 - \alpha)\tau$. Raising α makes z less sensitive to τ , so the return-rate response to the tax is weaker. At $\tau = 0$ that shows up as the additive term $\alpha r'(p)(p + h)$ on top of the companion factor μ_p . If $r' > 0$, a larger α raises the buyer factor and pulls ρ_τ^b toward $(1 - r)\rho_c$ from below (when $\mu_p < 1 - r$). If $r' < 0$, reclaim works in the opposite direction: softer tax pressure on z means less equilibrium screening, so the buyer factor falls as α rises. In either case $\alpha = 1$ forces both regimes to $(1 - r)\rho_c$, which is why full reclaim restores remittance neutrality for pass-through.

The companion note also signs the classical bias for ρ_c : the no-return formula overestimates pass-through if and only if $\mu_p > 1$. That comparison does not involve α . At $\tau = 0$ the list-price slope is $\mu_p = 1 - r(p) - r'(p)(p + h)$ under either remittance regime, because reclaim enters only through terms in $\alpha\tau$ that drop out of $\partial\mu/\partial p$. Production-cost pass-through is $\rho_c = S'(\mu)/(S'(\mu)\mu_p - D'(p))$, again with no α . The classical formula sets $\mu_p = 1$ in that expression, so whether it over- or underestimates ρ_c is pinned entirely by whether $\mu_p \gtrless 1$ —the same $\varepsilon_r(p) < -p/(p + h)$ cut as in the companion note, for every $\alpha \in [0, 1]$. Full reclaim is no exception: it equalizes the two tax rates at $(1 - r)\rho_c$ but leaves ρ_c and its bias untouched.

Result 3 (Pass-through bias under partial reclaim). *The companion bias cut for ρ_c relative to the classical no-return formula is unchanged by α , including at full reclaim $\alpha = 1$. Tax pass-through inherits that bias only through the reclaim factors in (8).*

5 Revenue and first-order surplus

On each gross sale the treasury collects τ . On each return it refunds $\alpha\tau$ to the remitter. Net revenue is therefore

$$G = \tau Q(1 - \alpha r^*), \quad (9)$$

where $r^* = r(p)$ under seller remittance and $r^* = r(p - (1 - \alpha)\tau)$ under buyer remittance. At $\tau = 0$ both give $dG/d\tau = Q(1 - \alpha r(p))$.

Private surplus $W = CS + PS$ is differentiated as in the companion note, using $dCS/d\tau = -Q(1 - r)\rho_\tau$ and $dPS/d\tau = Q(\mu_p\rho_\tau + \mu_\tau)$ at $\tau = 0$. Substituting (7)–(8) and writing $\Lambda \equiv 1 + r'(p)(p + h)\rho_c$ yields

$$\frac{dW}{d\tau^s} = -Q(1 - \alpha r(p)) \Lambda, \quad \frac{dW}{d\tau^b} = -Q(\mu_p + \alpha r'(p)(p + h)) \Lambda. \quad (10)$$

Social surplus $V = W + G$ then satisfies, at $\tau = 0$,

$$\frac{dV}{d\tau^s} = -Q(1 - \alpha r(p)) (\Lambda - 1), \quad \frac{dV}{d\tau^b} = Q[(1 - \alpha r(p)) - (\mu_p + \alpha r'(p)(p + h)) \Lambda]. \quad (11)$$

When $\alpha = 0$, (10)–(11) reproduce the companion note. When $\alpha \rightarrow 1$, seller remittance has $dW/d\tau^s \rightarrow -Q(1 - r)\Lambda$ and $dV/d\tau^s \rightarrow -Q(1 - r)(\Lambda - 1)$, so the unclaimed tax on returned units vanishes from the first-order social wedge and only the kept-unit share $1 - r$ remains in the private loss. Incomplete reclaim $\alpha < 1$ leaves a strictly larger seller-side resource cost $1 - \alpha r > 1 - r$ whenever $r \in (0, 1)$.

Result 4 (Partial recovery nests zero and full reclaim). *For $\alpha \in [0, 1]$, equations (1)–(11) nest the zero-reclaim companion note at $\alpha = 0$. At $\alpha = 1$, tax neutrality is restored and the seller-tax pass-through factor collapses to the keep rate $1 - r(p)$.*